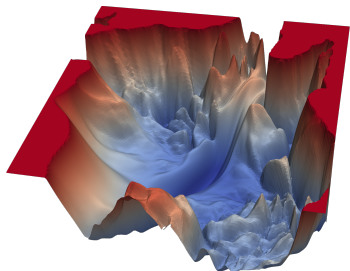
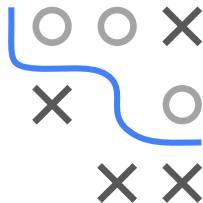


# Introduction to Machine Learning

## Advanced Risk Minimization Properties of Loss Functions



### Learning goals

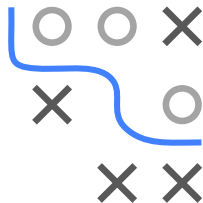
- Statistical properties
- Robustness
- Optimization properties
- Some fundamental terminology

# THE ROLE OF LOSS FUNCTIONS

- Should be designed to measure errors appropriately
- **Statistical** properties: choice of loss implies statistical assumptions about the distribution of  $y \mid \mathbf{x} = \tilde{\mathbf{x}}$   
(see *maximum likelihood vs. empirical risk minimization*)
- **Robustness** properties:  
some losses more robust towards outliers than others
- **Optimization** properties: computational complexity of

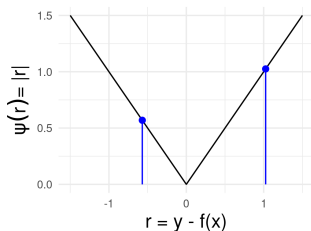
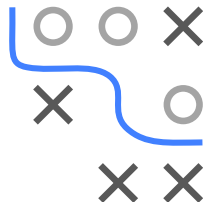
$$\arg \min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta)$$

is influenced by choice of the loss

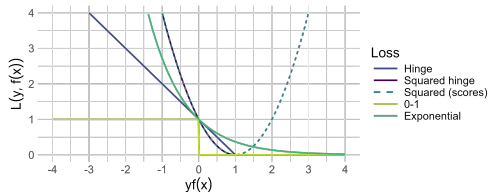


# LOSSES WITH ONE ARGUMENT

- Regr. losses often only depend on **residuals**  $r(\mathbf{x}) := y - f(\mathbf{x})$
- Classif. losses usually in terms of **margin**:  $\nu(\mathbf{x}) := y \cdot f(\mathbf{x})$



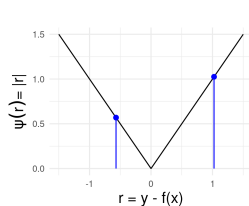
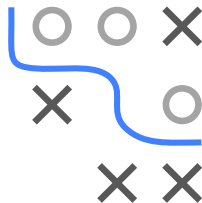
Distance-based:  $L_1$



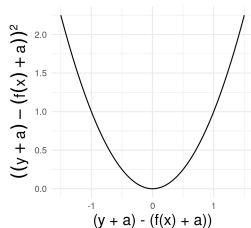
# SOME BASIC PROPERTIES

A loss is

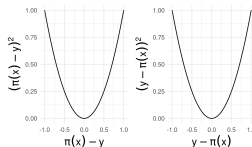
- **symmetric** if  $L(y, f(\mathbf{x})) = L(f(\mathbf{x}), y)$
- **translation-invariant** if  $L(y + a, f(\mathbf{x}) + a) = L(y, f(\mathbf{x}))$ ,  $a \in \mathbb{R}$
- **distance-based** if it can be written in terms of residual  
 $L(y, f(\mathbf{x})) = \psi(r)$  for some  $\psi : \mathbb{R} \rightarrow \mathbb{R}$ , and  $\psi(r) = 0 \Leftrightarrow r = 0$



Distance-based:  $L_1$



Transl.-invariant.:  $L_2$



Symmetric: Brier score

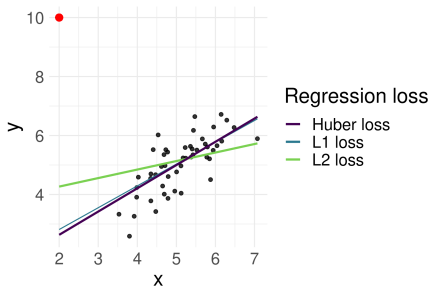
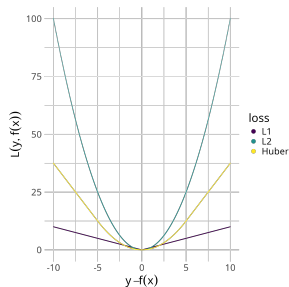
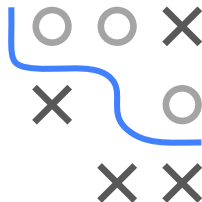
# ROBUSTNESS

Outliers (in  $y$ ) have large residuals  $r(\mathbf{x}) = y - f(\mathbf{x})$ . Some losses are more affected by large residuals than others. If loss goes up superlinearly (e.g.  $L2$ ) it is not robust, linear ( $L1$ ) or even sublinear losses are more robust.

| $y - f(\mathbf{x})$ | $L1$ | $L2$ | Huber ( $\epsilon = 5$ ) |
|---------------------|------|------|--------------------------|
| 1                   | 1    | 1    | 0.5                      |
| 5                   | 5    | 25   | 12.5                     |
| 10                  | 10   | 100  | 37.5                     |
| 50                  | 50   | 2500 | 237.5                    |

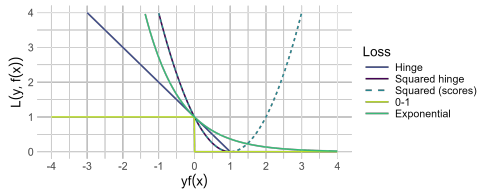
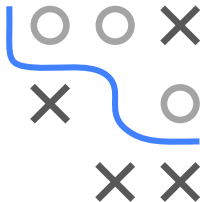
As a consequence, a model is less influenced by outliers than by “inliers” if the loss is **robust**.

Outliers e.g. strongly influence  $L2$ .



# OPTIMIZATION PROPERTIES: SMOOTHNESS

- Measured by number of continuous derivatives.  $f \in \mathcal{C}^k$  for  $k$ -times cont. differentiable.  $f$  and smooth for  $f \in \mathcal{C}^\infty$
- Usually want to have at least gradients in optimization of  $\mathcal{R}_{\text{emp}}(\theta)$
- If loss is not differentiable, might have to use derivative-free optimization (or worse, in case of 0-1)
- Smoothness of  $\mathcal{R}_{\text{emp}}(\theta)$  not only depends on  $L$ , but also requires smoothness of  $f(\mathbf{x})$ !



Squared, exponential and squared hinge losses are

continuously differentiable.

Hinge loss is continuous but not differentiable. 0-1 loss is not even continuous.

# OPTIMIZATION PROPERTIES: CONVEXITY

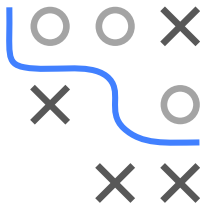
- $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$  is convex if

$$\mathcal{R}_{\text{emp}}(t \cdot \boldsymbol{\theta} + (1 - t) \cdot \tilde{\boldsymbol{\theta}}) \leq t \cdot \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + (1 - t) \cdot \mathcal{R}_{\text{emp}}(\tilde{\boldsymbol{\theta}})$$

$$\forall t \in [0, 1], \boldsymbol{\theta}, \tilde{\boldsymbol{\theta}} \in \Theta$$

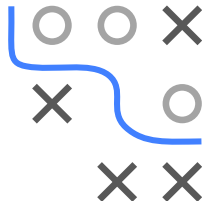
(strictly convex if above holds with strict inequality)

- In optimization, convex problems have several convenient properties, e.g. all local minima are global
- Strictly convex function has at most **one** global min (uniqueness)
- For  $\mathcal{R}_{\text{emp}} \in \mathcal{C}^2$ ,  $\mathcal{R}_{\text{emp}}$  is convex iff Hessian  $\nabla^2 \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$  is psd
- Above holds for arbitrary functions, not only risks

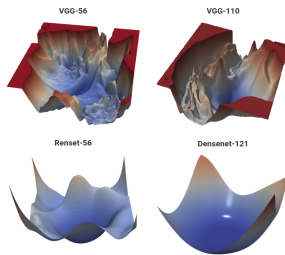


# OPTIMIZATION PROPERTIES: CONVEXITY

- Convexity of  $\mathcal{R}_{\text{emp}}(\theta)$  depends both on convexity of  $L(\cdot)$  (given in most cases) and  $f(\mathbf{x} \mid \theta)$  (often problematic)
- If we model our data using an exponential family distribution, we always get convex losses [▶ Wedderburn 1976](#)
- For  $f(\mathbf{x} \mid \theta)$  linear in  $\theta$ , linear/logistic/softmax/poisson/. . . regression are convex problems (all GLMs)!



The problem on the bottom right is convex, the others are not (note that very high-dimensional surfaces are coerced into 3D here).



[▶ Click for source](#)